

1. A) $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$f(x, y, z) = (-x + 3y - z, -3x + 5y - z, -3x + 3y + z)$$

a) $A_f = \begin{pmatrix} -1 & 3 & -1 \\ -3 & 5 & -1 \\ -3 & 3 & 1 \end{pmatrix} \in M_3(\mathbb{R})$

$\begin{matrix} \text{"} & \text{"} & \text{"} \\ f(e_1) & f(e_2) & f(e_3) \end{matrix}$

b) $p(\lambda) = \det(A_f - \lambda I_3) = \begin{vmatrix} -1-\lambda & 3 & -1 \\ -3 & 5-\lambda & -1 \\ -3 & 3 & 1-\lambda \end{vmatrix} =$

$$= -\lambda^3 + 5\lambda^2 - 8\lambda + 4 = -\lambda^3 + 2\lambda^2 + 3\lambda^2 - 6\lambda - 2\lambda + 4 =$$

$$= -\lambda^2(\lambda - 2) + 3\lambda(\lambda - 2) - 2(\lambda - 2) =$$

$$= (\lambda - 2)(-\lambda^2 + 3\lambda - 2) = -(\lambda - 2)(\lambda - 2)^2$$

$$p(\lambda) = 0 \Leftrightarrow -(\lambda - 2)(\lambda - 2)^2 = 0 \Rightarrow$$

$\Rightarrow \begin{cases} \lambda_1 = 1 \\ \lambda_2 = 2 \end{cases} \in \mathbb{R} \quad \text{Spec}(f) = \{1, 2\}$

$$m_u(\lambda_1) = 1$$

$$m_u(\lambda_2) = 2$$

Luizati proprii:

$$S_{\lambda_1=1} = \begin{cases} -2x + 3y - z = 0 \\ -3x + 4y - z = 0 \\ -3x + 3y = 0 \end{cases}$$

$$\Delta_A = \begin{vmatrix} -2 & 3 \\ -3 & 4 \end{vmatrix} = 1 \neq 0 \Rightarrow \text{rg}(A - \lambda_1 I_2) = 2 \Rightarrow$$

$$\Rightarrow \begin{cases} x, y \text{ are free} \\ z = x \end{cases}$$

$$\begin{cases} 3y = x + 2x \\ 3y = 3x \end{cases} \Rightarrow \begin{cases} x = x \\ y = x \end{cases}$$

$$\text{deci } v_{\lambda_1} = \{ x(1, 1, 1) / x \in \mathbb{R} \} \Rightarrow v_{\lambda_1} = \text{span}\{v_1\}$$

$$S_{\lambda_2} = 2 \begin{cases} -3x + 3y - z = 0 \\ -3x + 3y - z = 0 \\ -3x + 3y - z = 0 \end{cases} \Rightarrow \begin{cases} \text{rg}(A - \lambda_2 I_3) = 1 \\ -3x + 3y - z = 0 \\ z = -3x + 3y \end{cases}$$

$$\begin{cases} z \text{ free} \\ x = m \\ y = r \end{cases} \Rightarrow z = -3m + 3r$$

$$\text{deci } v_{\lambda_2} = \{ m(1, 0, -3) + r(0, 1, 3), m, r \in \mathbb{R} \}$$

$$v_{\lambda_2} = \text{span}\{v_2, v_3\}$$

c) stud de este diag

$$v_{\lambda_1} = \text{span}\{v_1\} \Rightarrow \dim v_{\lambda_1} = 1 = m_a(\lambda_1)$$

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -3 & 3 \end{pmatrix} \Rightarrow \Delta_A = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \Rightarrow \text{rg}(A - \lambda_1 I_2) = 2 \Rightarrow$$

$$v_{\lambda_2} = \text{span}\{v_2, v_3\} \Rightarrow \dim v_{\lambda_2} = 2 = m_a(\lambda_2)$$

$$d) D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, P = \{ v_1 = (1, 1, 1), v_2 = (1, 0, -3), v_3 = (0, 1, 3) \}$$

$$\text{ii) } f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$f(x, y, z) = (6x - 5y - 3z, 3x - 2y - 2z, 2x - 2z)$$

$$a) A_f = \begin{pmatrix} 6 & -5 & -3 \\ 3 & -2 & -2 \\ 2 & 0 & -2 \end{pmatrix} \in M_3(\mathbb{R})$$

$$b) p(x) = \det(A_f - x I_3) = \begin{vmatrix} 6-x & -5 & -3 \\ 3 & -2-x & -2 \\ 2 & 0 & -2-x \end{vmatrix} =$$

$$= -x^3 + 4x^2 - 5x + 2 = -x^3 + x^2 + 3x^2 - 3x + 2x + 2 =$$

$$= -x^2(x-1) + 3x(x-1) - 2(x-1) = -(x-1)^2(x-2)$$

$$p(x) = 0 \Rightarrow \begin{cases} x_1 = 1 & ; \text{ m.a. } (x_1) = 2 \\ x_2 = 2 & ; \text{ m.a. } (x_2) = 1 \end{cases} \quad \text{Spec} = \{1, 2\}$$

Subspații proprii:

$$\text{p.e. } x_1 = 1 \quad \begin{cases} 5x - 5y - 3z = 0 \\ 3x - 3y - 2z = 0 \\ 2x - 2y - z = 0 \end{cases}$$

$$A_x = \begin{vmatrix} -3 & -2 \\ -2 & -1 \end{vmatrix} = 1 \neq 0 \Rightarrow \text{rg}(A_f - x_1 I_3) = 2$$

$$\begin{cases} y, z \text{ nec. } \text{p.} \\ x = \alpha \text{ nec. } \text{p.} \end{cases}$$

$$\begin{cases} 3y + 2z = 3\alpha & \Rightarrow y = \alpha & \Rightarrow \\ 2y + z = 2\alpha & z = 0 \end{cases}$$

$$\text{deci } V_{x_1} = \{ \alpha (1, 1, 0) / \alpha \in \mathbb{R} \}$$

$$S_{\lambda_2=2} \begin{cases} 4x - 5y - 3z = 0 \\ 3x - 4y - 2z = 0 \\ 2x - 2y - 2z = 0 \end{cases}$$

$$A_{\lambda_2} = \begin{pmatrix} 4 & -5 \\ 3 & -4 \end{pmatrix} = -1 \neq 0 \rightarrow \text{rg}(A_{\lambda_2} - \lambda_2 I_2) = 2$$

$$\begin{cases} x, y \text{ free} \\ z = 0 \end{cases}$$

$$\begin{cases} 4x - 5y = 3z \\ 3x - 4y = 2z \end{cases} \Rightarrow \begin{aligned} -y &= -7x \Rightarrow y = 7x \\ x &= 8x \end{aligned} \quad \begin{aligned} z &= 13 \\ x &= 213 \end{aligned}$$

deci

$$V_{\lambda_2} = \{ p(2, 1, 1) \mid p \in \mathbb{R} \}$$

e) stud diag

$$V_{\lambda_1} = \text{span}\{v_1\} \Rightarrow \dim V_{\lambda_1} = 1 \neq m_A(\lambda_1) = 2$$

\(\Rightarrow\) matrice \(A\) nu este diag

71 $\mathcal{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^4$

$$\mathcal{A}(x, y, z, t) = (x + y + z + t, x + y - z - t, x - y + z - t, x - y - z + t)$$

$$A_{\mathcal{A}} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \in M_4(\mathbb{R})$$

$$\mathcal{A}(e_1) \quad \mathcal{A}(e_2) \quad \mathcal{A}(e_3) \quad \mathcal{A}(e_4)$$

$$b) \quad p(\lambda) = \det(A - \lambda I_4) = \begin{vmatrix} 1-\lambda & 1 & 1 & 1 \\ 1 & 1-\lambda & -1 & -1 \\ 1 & -1 & 1-\lambda & -1 \\ 1 & -1 & -1 & 1-\lambda \end{vmatrix} =$$

$$= \lambda^4 - 4\lambda^3 + 16\lambda^2 - 16 = \lambda^4 + 2\lambda^3 - 6\lambda^3 + 12\lambda^2 + 12\lambda^2 + 24\lambda - 8\lambda - 16 =$$

$$= \lambda^3(\lambda+2) - 6\lambda^2(\lambda+2) + 12\lambda(\lambda+2) - 8(\lambda+2) =$$

$$= (\lambda+2)(\lambda^3 - 6\lambda^2 + 12\lambda - 8) = (\lambda+2)(\lambda-2)^3$$

$$p(\lambda) = 0 \Leftrightarrow \begin{cases} \lambda_1 = -2, & m_a(\lambda_1) = 1 \\ \lambda_2 = 2, & m_a(\lambda_2) = 3 \end{cases} \quad \text{Spec} = \{-2, 2\}$$

$$S_{\lambda_1 = -2} = \begin{cases} 3x + y + z + t = 0 \\ x + 3y - z - t = 0 \\ x - y + 3z - t = 0 \\ x - y - z + 3t = 0 \end{cases}$$

$$A_{\lambda_1} = \begin{vmatrix} 3 & 1 & 1 \\ 1 & 3 & -1 \\ 1 & -1 & 3 \end{vmatrix} = 16 \neq 0 \Rightarrow \text{rang}(A - \lambda_1 I_4) = 3$$

$$\begin{cases} x, y, z \text{ free} \\ t = \alpha \text{ free} \end{cases}$$

$$\begin{cases} 3x + y + z = -\alpha \\ x + 3y - z = \alpha \\ x - y + 3z = \alpha \end{cases} \Rightarrow \begin{cases} 4x + 4y = 0 \\ 4x = -4y \\ x = -y \\ y = z \end{cases} \Rightarrow \begin{cases} x = -\alpha \\ y = \alpha \\ z = \alpha \end{cases}$$

$$\text{dec} \quad v_{\lambda_1} = \{ \alpha(-1, 1, 1, 1) / \alpha \in \mathbb{R} \}$$

$$5 \lambda_2 = 2 \begin{cases} -x + y + z + t = 0 & 1-1 \\ x - y - z - t = 0 \\ x - y - z - t = 0 \\ x - y - z - t = 0 \end{cases} \Rightarrow \begin{cases} x - y - z - t = 0 \\ \text{rg}(A - \lambda_2 I_4) = 1 \end{cases}$$

$$\begin{cases} x \text{ rec } \text{rec} \\ y = m, z = r, t = \theta \text{ rec rec} \end{cases} \Rightarrow x = m + r + \theta$$

$$\Rightarrow \text{dec } v_{x_2} = \{ m \underset{v_2}{(1, 1, 0, 0)} + r \underset{v_3}{(1, 0, 1, 0)} + \theta \underset{v_4}{(1, 0, 0, 1)} \} \\ m, r, \theta \in \mathbb{R}$$

c) stud diag

$$v_{x_1} = \text{sp} \{ v_1 \} \Rightarrow \dim v_{x_1} = 1 = m a (\lambda_1)$$

$$X = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow \Delta_X = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = 1 \neq 0 \Rightarrow \text{rg } X = 3 \Rightarrow$$

$$v_{x_2} = \text{sp} \{ v_2, v_3, v_4 \} \Rightarrow \dim v_{x_2} = 3 = m a (\lambda_2)$$

-) A not diag

$$d) D = \begin{pmatrix} -2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

$$B = \{ v_1 = (-1, 1, 1, 1), v_2 = (1, 1, 0, 0), v_3 = (1, 0, 1, 0),$$

$$v_4 = (1, 0, 0, 1) \}$$

$$2. a) \quad A = \begin{pmatrix} 0 & 4 \\ 2 & 2 \end{pmatrix}$$

$$p(x) = \det(A - x I_2) = \begin{vmatrix} -x & 4 \\ 2 & 2-x \end{vmatrix} = x^2 - 2x - 8 = (x-4)(x+2)$$

$$p(x) = 0 \Rightarrow \begin{cases} x_1 = -2 & m_a(x_1) = 1 \\ x_2 = 4 & m_a(x_2) = 1 \end{cases} \quad \text{Spec } A = \{-2, 4\}$$

Subspații proprii:

$$S_{x_1=-2} \begin{cases} 2x + 4y = 0 \\ 2x + 4y = 0 \end{cases} \Rightarrow \begin{cases} 2x = -4y \\ x = -2y \end{cases} \quad \text{cu } (A - x I_2) = 0$$

$$\begin{cases} x \text{ nec } nr \\ y = x \text{ nec nec} \end{cases} \Rightarrow x = -2\alpha$$

$$\text{deci } v_{x_1} = \left\{ \alpha \begin{pmatrix} -2 \\ 1 \end{pmatrix} \mid \alpha \in \mathbb{R} \right\}$$

$$S_{x_2=4} \begin{cases} -4x + 4y = 0 \\ 2x - 2y = 0 \end{cases} \Rightarrow \begin{cases} 4x = 4y \\ x = y \end{cases}$$

$$\begin{cases} x \text{ nec } nr \\ y = \beta \text{ nec nec} \end{cases} \Rightarrow x = \beta$$

$$\text{deci } v_{x_2} = \left\{ \beta \begin{pmatrix} 1 \\ 1 \end{pmatrix} \mid \beta \in \mathbb{R} \right\}$$

$$b) \quad A = \begin{pmatrix} 3 & 4 \\ 0 & 5 \end{pmatrix}$$

$$p(x) = \det(A - x I_2) = \begin{vmatrix} 3-x & 4 \\ 0 & 5-x \end{vmatrix} = (x-5)(x-3)$$

$$p(x) = 0 \Rightarrow \begin{cases} x_1 = 3 & m_a(x_1) = 1 \\ x_2 = 5 & m_a(x_2) = 1 \end{cases}$$

$$S'_{\lambda_1=3} \begin{cases} 4y = 0 \\ 2y = 0 \end{cases} \Rightarrow \begin{matrix} y = 0 \\ \text{rang}(A - \lambda_1 I_3) = 1 \end{matrix}$$

$$\begin{cases} x = \alpha \text{ rec } \cancel{\text{rec}} \text{ rec} \Rightarrow \\ y = \text{rec } m \end{cases}$$

$$\text{decu: } V_{\lambda_1} = \{ \alpha (1, 0) \mid \alpha \in \mathbb{R} \}$$

$\uparrow$
 v_1

$$S'_{\lambda_2=5} \begin{cases} -2x + 4y = 0 \\ 0 + 0 = 0 \end{cases} \Rightarrow x = 2y$$

$$\begin{cases} x = \text{rec } m \\ y = \beta \text{ rec } m \end{cases} \Rightarrow x = 2\beta$$

$$\text{decu } \cancel{V_{\lambda_2}} = V_{\lambda_2} = \{ \beta (2, 1) \mid \beta \in \mathbb{R} \}$$

$\uparrow$
 v_2

$$1) \quad A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix}$$

$$p(x) = \det(A - xI_3) = \begin{vmatrix} 1-x & 0 & 0 \\ 0 & -x & 2 \\ 0 & 2 & -x \end{vmatrix} =$$

$$= -x^3 + x^2 + 4x - 4 = -x^2(x-1) + 4(x-1) = (-x^2+4)(x-1) =$$

$$= -(x^2-4)(x-1) = -(x+2)(x-2)(x-1)$$

$$p(x) = 0 \Leftrightarrow \begin{cases} \lambda_1 = -2 & , & m_a(\lambda_1) = 1 \\ \lambda_2 = 2 & , & m_a(\lambda_2) = 1 \\ \lambda_3 = 1 & , & m_a(\lambda_3) = 1 \end{cases}$$

Substitution procedure:

$$\begin{cases} \lambda_1 = -2 \\ \begin{cases} x = 0 \\ 2y + 2z = 0 \\ 2y + 2z = 0 \end{cases} \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = -z \end{cases}$$

$$A_{\lambda} = \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} = 2 \neq 0 \Rightarrow \text{rg}(A - \lambda_1 I_2) = 2$$

$$\begin{cases} x, y \text{ free var} \\ z = \alpha \text{ free var} \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = -\alpha \end{cases}$$

$$\text{dec } v_{\lambda_1} = \{ \alpha (0, -1, 1) \mid \alpha \in \mathbb{R} \}$$

$\uparrow$
 v_1

$$\begin{cases} \lambda_2 = 2 \\ \begin{cases} -x = 0 \\ -2y + 2z = 0 \\ 2y - 2z = 0 \end{cases} \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = z \end{cases}$$

$$A_{\lambda} = \begin{vmatrix} -1 & 0 \\ 0 & -2 \end{vmatrix} = 2 \neq 0 \Rightarrow \text{rg}(A - \lambda_2 I_2) = 2$$

$$\begin{cases} x, y \text{ free var} \\ z = \beta \text{ free var} \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = \beta \end{cases}$$

$$\text{dec } v_{\lambda_2} = \{ \beta (0, 1, 1) \mid \beta \in \mathbb{R} \}$$

$\uparrow$
 v_2

$$\begin{cases} \lambda_3 = 1 \\ \begin{cases} 0 = 0 \\ -y + 2z = 0 \\ 2y - z = 0 \end{cases} \end{cases} \Rightarrow \begin{cases} y = 2z \\ z = 2y \end{cases} \Rightarrow \begin{cases} y = 0 \\ z = 0 \end{cases}$$

$$A_{\lambda} = \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} = -3 \neq 0 \Rightarrow \text{rg}(A - \lambda_3 I_2) = 2$$

$$\begin{cases} y, z \text{ free var} \\ x = 0 \text{ free var} \end{cases} \Rightarrow \begin{cases} y = 0 \\ z = 0 \end{cases}$$

$$\text{dec } v_{\lambda_3} = \{ \alpha (1, 0, 0) \mid \alpha \in \mathbb{R} \}$$

$\uparrow$
 v_3

$$d) A = \begin{pmatrix} 0 & 3 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$p(x) = \det(A - xI_3) = \begin{vmatrix} -x & 3 & 0 \\ 4 & -x & 0 \\ 0 & 0 & 3-x \end{vmatrix} = -x^3 + 3x^2 + 36x - 108$$

$$= -x^2(x-3) + 36(x-3) = -(x-36)(x-3) = -(x+6)(x-6)(x-3)$$

$$p(x) = 0 \Leftrightarrow \begin{cases} x_1 = -6 & m_a(x_1) = 1 \\ x_2 = 6 & m_a(x_2) = 1 \\ x_3 = 3 & m_a(x_3) = 1 \end{cases}$$

Subspații proprii

$$S_{x_1 = -6} \begin{cases} 6x + 3y = 0 \\ 4x + 6y = 0 \\ 3z = 0 \end{cases} \Rightarrow \begin{matrix} z = 0 \\ x = -\frac{3}{2}y \Rightarrow y = -\frac{2}{3}x \end{matrix}$$

$$A_{x_1} = \begin{vmatrix} 6 & 0 \\ 0 & 3 \end{vmatrix} = 18 \neq 0 \Rightarrow \dim(A - x_1 I_2) = 2$$

$$\begin{cases} y, z \text{ nec. } 0 \\ x = \alpha \end{cases} \Rightarrow \begin{matrix} z = 0 \\ y = -\frac{2}{3}\alpha \end{matrix}$$

$$\text{deci } v_{x_1} = \left\{ \alpha \left(1, -\frac{2}{3}, 0 \right) \mid \alpha \in \mathbb{R} \right\}$$

$$S_{x_2 = 6} \begin{cases} -6x + 3y = 0 \\ 4x - 6y = 0 \\ -3z = 0 \end{cases} \Rightarrow \begin{matrix} z = 0 \\ y = \frac{2}{3}x \end{matrix}$$

$$A_{x_2} = \begin{vmatrix} -6 & 0 \\ 0 & -3 \end{vmatrix} = 18 \neq 0 \Rightarrow \dim(A - x_2 I_2) = 2$$

$$\begin{cases} z, z \text{ rec pr} \\ x = \beta \end{cases} \Rightarrow \begin{cases} z = 0 \\ y = \frac{2}{-3} \beta \end{cases} \Rightarrow$$

$$\text{dec } v_{x_2} = \left\{ \beta \left(1, \frac{2}{3}, 0 \right) \mid \beta \in \mathbb{R} \right\}$$

$\uparrow$
 v_2

$$y_{x_3} = 3 \begin{cases} -3x + 3y = 0 \\ x - 3y = 0 \\ 0 = 0 \end{cases} \Rightarrow \begin{cases} x = 3y \\ y = \frac{1}{3}x \end{cases} \Rightarrow x = y = 0$$

$$\Delta x = \begin{vmatrix} -3 & 3 \\ 1 & -3 \end{vmatrix} = -2 \neq 0 \Rightarrow \text{rg}(A - x_3 I_3) = 2$$

$$\begin{cases} x, y \text{ rec pr} \\ z = \gamma \text{ rec rec} \end{cases} \Rightarrow \begin{cases} x = z = 0 \end{cases} \Rightarrow$$

$$\text{dec } v_{x_3} = \left\{ \gamma (0, 0, 1) \mid \gamma \in \mathbb{R} \right\}$$

$\uparrow$
 v_3

3. a) $v_{x_1} = \eta \langle v_1 \rangle \Rightarrow \dim v_{x_1} = 1 = m a(x_1)$

$v_{x_2} = \eta \langle v_2 \rangle \Rightarrow \dim v_{x_2} = 1 = m a(x_2)$

$\Rightarrow$ A diag

$$D = \begin{pmatrix} -2 & 0 \\ 0 & 4 \end{pmatrix}$$

b) $v_{x_1} = \eta \langle v_1 \rangle \Rightarrow \dim v_{x_1} = 1 = m a(x_1)$

$v_{x_2} = \eta \langle v_2 \rangle \Rightarrow \dim v_{x_2} = 1 = m a(x_2)$

$\Rightarrow$ A diag

$$D = \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}$$

$$c) \quad V_{\lambda_1} = \ker \langle v_1 \rangle \Rightarrow \dim V_{\lambda_1} = 1 = m_a(\lambda_1)$$

$$V_{\lambda_2} = \ker \langle v_2 \rangle \Rightarrow \dim V_{\lambda_2} = 1 = m_a(\lambda_2)$$

$$V_{\lambda_3} = \ker \langle v_3 \rangle \Rightarrow \dim V_{\lambda_3} = 1 = m_a(\lambda_3)$$

$\Rightarrow$ A diag

$$D = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$d) \quad V_{\lambda_1} = \ker \langle v_1 \rangle \Rightarrow \dim V_{\lambda_1} = 1 = m_a(\lambda_1)$$

$$V_{\lambda_2} = \ker \langle v_2 \rangle \Rightarrow \dim V_{\lambda_2} = 1 = m_a(\lambda_2)$$

$$V_{\lambda_3} = \ker \langle v_3 \rangle \Rightarrow \dim V_{\lambda_3} = 1 = m_a(\lambda_3)$$

$\Rightarrow$ A diag

$$D = \begin{pmatrix} -6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

Subspațiile proprii:

$$S_{\lambda_1 = -1} \begin{cases} 2x + 4y = 0 \\ -3y + 3z = 0 \\ y + z = 0 \end{cases} \Rightarrow \begin{cases} 2x + 4y = 0 \\ y + z = 0 \end{cases} \Rightarrow \begin{cases} x = -2y \\ y = -z \end{cases}$$

$$\Delta x = \begin{vmatrix} 2 & 4 \\ 1 & 1 \end{vmatrix} = -2 \neq 0 \Rightarrow \text{rang}(A_T - \lambda_1 I_3) = 2$$

$$\begin{cases} x, y \text{ nec. pr.} \\ z = \alpha \text{ nec. rec.} \end{cases} \Rightarrow \begin{cases} x = -2\alpha \\ y = -\alpha \end{cases}$$

$$\text{deci } v_{\lambda_1} = \left\{ \alpha \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} \mid \alpha \in \mathbb{R} \right\}$$

$$S_{\lambda_2 = 1} \begin{cases} 4y = 0 \\ y + 3z = 0 \\ y - z = 0 \end{cases} \Rightarrow \begin{cases} y = 0 \\ z = 0 \end{cases}$$

$$\Delta x = \begin{vmatrix} 4 & 0 \\ 1 & -1 \end{vmatrix} = -4 \neq 0 \Rightarrow \text{rang}(A_T - \lambda_2 I_3) = 2$$

$$\begin{cases} y, z \text{ nec. pr.} \\ x = \beta \text{ nec. rec.} \end{cases} \Rightarrow$$

$$\text{deci } v_{\lambda_2} = \left\{ \beta \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \mid \beta \in \mathbb{R} \right\}$$

$$S_{\lambda_3 = 3} \begin{cases} -2x + 4y = 0 \\ -y + 3z = 0 \\ y - 3z = 0 \end{cases} \Rightarrow \begin{cases} x = 2y \\ y = 3z \end{cases}$$

$$\Delta x = \begin{vmatrix} -2 & 4 \\ -1 & 3 \end{vmatrix} = -2 \neq 0 \Rightarrow \text{rang}(A_T - \lambda_3 I_3) = 2$$

$$\begin{cases} x, y \text{ nec. pr.} \\ z = \gamma \text{ nec. rec.} \end{cases} \Rightarrow \begin{cases} x = 6\gamma \\ y = 3\gamma \end{cases}$$

$$\text{deci } v_{\lambda_3} = \left\{ \gamma \begin{pmatrix} 6 \\ 3 \\ 1 \end{pmatrix} \mid \gamma \in \mathbb{R} \right\}$$

$$4. \quad T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(x, y, z) = (x + 4y, 2y + 3z, y)$$

$$a) \quad T(x) = Ax, \text{ unde } x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \text{ in } A = \begin{pmatrix} 1 & 4 & 0 \\ 0 & 2 & 3 \\ 0 & 1 & 0 \end{pmatrix} \in M_3(\mathbb{R})$$

$$\text{fie } x_1, x_2 \in \mathbb{R}^3$$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \quad \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$$

$$\text{in } x_1, x_2 \in \mathbb{R}$$

$$\begin{aligned} T(x_1 x_2 + x_2 x_2) &= A(x_1 x_1 + x_2 x_2) = A(x_1 x_1) + A(x_2 x_2) = \\ &= (Ax_1) x_1 + (Ax_2) x_2 = (x_1 A) x_1 + (x_2 A) x_2 = x_1 (Ax_1) + x_2 (Ax_2) = \\ &= x_1 T(x_1) + x_2 T(x_2) \Rightarrow T \text{ apl lin} \end{aligned}$$

$$b) \quad A_T = \begin{pmatrix} 1 & 4 & 0 \\ 0 & 2 & 3 \\ 0 & 1 & 0 \end{pmatrix} \in M_3(\mathbb{R})$$

$$c) \quad p(\lambda) = \det(A_T - \lambda I_3) = \begin{vmatrix} 1-\lambda & 4 & 0 \\ 0 & 2-\lambda & 3 \\ 0 & 1 & -\lambda \end{vmatrix} =$$

$$= -\lambda^3 + 3\lambda^2 + \lambda - 3 = -\lambda^2(\lambda - 3) + \lambda - 3 = -(\lambda^2 - 1)(\lambda - 3)$$

$$= -(\lambda + 1)(\lambda - 1)(\lambda - 3)$$

$$p(\lambda) = 0 \Leftrightarrow$$

$$\begin{cases} \lambda_1 = -1 & m_a(\lambda_1) = 1 \\ \lambda_2 = 1 & m_a(\lambda_2) = 1 \\ \lambda_3 = 3 & m_a(\lambda_3) = 1 \end{cases}$$

$$S_{\text{spec}} = \{-1, 1, 3\}$$

$$d) \quad v_{\lambda_1} = \eta \langle v_1 \rangle \Rightarrow \dim v_{\lambda_1} = 1 = m a(\lambda_1)$$

$$v_{\lambda_2} = \eta \langle v_2 \rangle \Rightarrow \dim v_{\lambda_2} = 1 = m a(\lambda_2)$$

$$v_{\lambda_3} = \eta \langle v_3 \rangle \Rightarrow \dim v_{\lambda_3} = 1 = m a(\lambda_3)$$

$\Rightarrow A_T$ diag

$$e) \quad D = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$B = \{ v_1 = (2, -1, 1), v_2 = (1, 0, 0), v_3 = (6, 3, 1) \}$$

$$C = (v_1 \ v_2 \ v_3) = \begin{pmatrix} 2 & 1 & 6 \\ -1 & 0 & 3 \\ 1 & 0 & 1 \end{pmatrix}$$